

Behavior Safety of Autonomous Interactive Agents: Risks, Attacks, Defenses, Evaluation, and Governance

XINJIE LAI, School of Communication and Information Engineering, Shanghai, Shanghai University, China

WEIHAO ZUO, School of Communication and Information Engineering, Shanghai, Shanghai University, China

TIANJIE JU, School of Computer Science, Shanghai, Shanghai Jiao Tong University, China

HAODONG ZHAO, School of Computer Science, Shanghai, Shanghai Jiao Tong University, China

PENGZHOU CHENG[†], School of Computer Engineering and Science, Shanghai University, China

XINPENG ZHANG[†], School of Computer Engineering and Science, Shanghai University, China

CCS Concepts: • **Security and privacy** → **Software and application security**; • **Computing methodologies** → **Cognitive science**; **Natural language generation**.

Additional Key Words and Phrases: Autonomous Interactive Agents, Behavior Safety, Inherent Cognitive Vulnerability, Extrinsic Decision Uncontrollable, Defenses, Benchmark Evaluation

ACM Reference Format:

Xinjie Lai, Weihao Zuo, Tianjie Ju, Haodong Zhao, Pengzhou Cheng[†], and Xinpeng Zhang[†]. 2026. Behavior Safety of Autonomous Interactive Agents: Risks, Attacks, Defenses, Evaluation, and Governance. *Proc. ACM Meas. Anal. Comput. Syst.* 37, 4, Article 111 (August 2026), 24 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

Large Language Model agents have progressively revolutionized from passive reactive systems into autonomous interactive systems capable of proactively perceiving, independently planning and executing sophisticated tasks in digital or physical environments. This evolution is motivated by the explosive emergence of techniques in tool invocation, workflow optimization, and memory management. Specifically, these autonomous interactive agents can analyze user intentions, decompose and schedule long-horizon goals, and then act through diverse external tools, skills, or actuators. Within this closed feedback loop, agents continuously reflect on and improve their interactions with dynamic open environments under minimal human supervision, such as fixing local files and browsing the web.

[†] Corresponding authors.

Authors' Contact Information: Xinjie Lai, School of Communication and Information Engineering, Shanghai, Shanghai University, China, 1991064479@shu.edu.cn; Weihao Zuo, School of Communication and Information Engineering, Shanghai, Shanghai University, China, z2630109079@gmail.com; Tianjie Ju, School of Computer Science, Shanghai, Shanghai Jiao Tong University, China, jometeorie@sjtu.edu.cn; Haodong Zhao, School of Computer Science, Shanghai, Shanghai Jiao Tong University, China, zhaohadong@sjtu.edu.cn; Pengzhou Cheng[†], School of Computer Engineering and Science, Shanghai University, China, pengzhouchengai@gmail.com; Xinpeng Zhang[†], School of Computer Engineering and Science, Shanghai University, China, xzhang@shu.edu.cn.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, or post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

© 2026 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM 2476-1249/2026/8-ART111

<https://doi.org/XXXXXXX.XXXXXXX>

The marked shift from content generation to autonomous action consequently entails the safety issue of agents transcending content and escalating into a more threatening behavioral risk. The content safety concerns whether a model says anything incorrect or unauthorized. Behavioral safety, in contrast, concerns whether an agent performs any harmful action. Indeed, the inherent cognitive defects of agents—stemming from infrastructure design flaws, or other unnoticed inappropriate latent properties such as experience-following execution trajectories—will be amplified when subjected to malicious attacks. Furthermore, these internal vulnerabilities and external risks may give rise to erroneous execution, unauthorized invocation, privacy leakage, memory contamination, collective uncontrolled behavior, or even unexpected real-world harm.

Existing surveys of agent safety are predominantly confined to content output risks, attack surfaces, engineering capabilities, collaborative mechanisms, or model training, which still adhere to a model-centric paradigm. A safety framework through the lens of behavioral chain remains largely underexplored. To bridge the gap, we propose a unified behavior-centric safety framework encompassing the entire process of cognition, execution, environment, and collaboration.

The main contributions of this survey can be summarized as follows:

- We propose a unified definition and taxonomy of agent behavioral safety, bridging the gap between static content safety and dynamic behavioral safety under a coherent perspective.
-

This survey seeks to provide a thorough understanding of the behavior challenges posed by autonomous interactive agents and contribute to the development of safer and more trustful AGI systems. The rest of this article is structured as follows: Section 2

With the continued advancement of LLM-driven agentic systems, artificial intelligence is progressively transitioning from a traditional information generation tool into an autonomous entity capable of perceiving environments, planning tasks, invoking tools, and executing complex decision-making processes.

Yang et al. [113] introduces the notion of the Agentic Web as a novel paradigm for the future Internet, positing that the primary actors in cyberspace will no longer be limited to human users in the conventional sense, but will increasingly be AI agents endowed with autonomous reasoning, collaborative interaction, and persistent execution capabilities. This paradigm signals a profound architectural shift of the Internet from a Human-Centric Web toward an Agent-Centric Web, in which interaction paradigms, resource organization mechanisms, protocol structures, and even economic operating models must be redesigned around the autonomous behaviors of intelligent agents.

Departing from traditional perspectives that regard agents merely as external functional wrappers of LLMs, this work further conceptualizes agents as autonomous Internet entities, emphasizing that they are not only task executors but also independent participants embedded within future network ecosystems. On this basis, the paper systematically constructs an integrated framework for the Agentic Web along three dimensions—Intelligence, Interaction, and Economics—aiming to address the structural mismatch between existing Internet infrastructure and the large-scale deployment of AI agents across the web.

Agentic AI refers to an autonomous intelligent system characterized by persistent objectives. It is capable of perceiving the environment, conducting multi-step planning, and exerting tangible influence on external systems through tools or actuators. Furthermore, it can reflect on execution outcomes and dynamically adjust its internal state under clearly defined human supervision, safety policies, and operational constraints [73].

Su et al. [82] propose a hierarchical autonomy framework ranging from L1 to L5 to characterize the evolutionary trajectory of large-model agents in terms of decision-making independence and environmental control capability. At the L1 level, agents exhibit only passive response behaviors and limited tool invocation abilities. L2 introduces constrained memory mechanisms, enabling agents to perform task planning and execution within predefined workflows. At L3, agents further acquire cross-step decision-making capability and dynamic task adaptation.

L4 achieves long-term autonomous operation in open environments, allowing agents to autonomously invoke external tools and maintain persistent internal states. Finally, L5 represents highly autonomous agents capable of complex goal coordination and self-optimization with minimal human intervention. The paper further argues that, as the level of autonomy increases, the associated risks evolve from output-level errors toward system-level behavioral loss of control, privilege abuse, and autonomous offensive behaviors.

The evolutionary trajectory of GUI agents driven by LLM has undergone a significant paradigm shift, progressing from script-based automation to agent-centric transformation and finally to a state of intrinsic drive powered by large language models [119].

2 Background

3 Endogenous Cognitive Vulnerability

The taxonomy of endogenous cognitive vulnerabilities is organized according to the functional cognitive architecture of autonomous interactive agents, including foundational reasoning mechanisms, memory systems, tool-interaction modules, and multi-agent coordination structures.

3.1 Foundation Model Vulnerability

Autonomous interactive agents integrate advanced LLMs as their core internal engines capable of iteratively interacting with the environment, processing feedback, and making decisions in real-time [13]. Naturally, agents not only inherit the inherent security vulnerabilities from foundation models, but also suffer from emerging specific risks due to other introduced components. Foundation model safety vulnerabilities are categorized according to distinct failure modes emerging during autonomous cognitive processing, including behavioral objective deviation, intention preservation failure, heuristic reasoning shortcuts, recursive planning instability, and reflective rigidity.

3.1.1 Behavioral objective deviation. Behavioral objective deviation arises when the behaviors generated by foundation models fail to satisfy the operational objectives required for autonomous interactive agents. These deviations may compromise the truthfulness, reliability, fairness, or utility of agent behaviors, leading to unsafe or misaligned interactions with users and external environments. Prior research has primarily identified four representative manifestations of behavioral objective deviation: hallucinations, factual errors, inherent bias, and alignment tax.

- **Hallucination.** Hallucination refers to fabricated, logically inconsistent, or erroneous outputs, which are often spontaneously generated within the agent's own cognitive processes [51]. Unlike LLM hallucinations those are primarily confined to linguistic errors, agent hallucinations can emerge throughout the entire agent pipeline and propagate deviations along the long-horizon trajectory [55]. For example, planning hallucinations generate steps that cannot be executed in reality, tool hallucinations fabricate the return value of a non-existent function, and reflection hallucinations continue to frequently repeat the exact same errors [97], despite being capable of generating convincing self-criticism. In embodied or autonomous settings, these hallucinations may further lead to incorrect task execution, unsafe system operations, unreliable interactions in the real world, thus degrading overall reliability.

Behavioral Safety Constraints. Liang et al. [48] proposed a training framework named Reinforcement Learning from Knowledge Feedback (RLKF). By learning the association between hidden representations and answer correctness, the framework enabled the model to perceive uncertainty in its own reasoning and generation processes. Building on this intrinsic confidence estimation capability, the model can actively suppress unreliable outputs in high-stakes generative scenarios, thereby mitigating hallucinations. Gosmar et al. [23] constructed a multi-agent orchestration mechanism based on the OVON framework. During

multi-turn collaborative reasoning and agent interactions, the method continuously filtered and corrected potential errors through process-level constraints and natural language-based content recalibration. This mechanism reduces the propagation and accumulation of hallucinations across the generation pipeline by enforcing structured control over intermediate reasoning and generated content. Furthermore, Pan et al. [72] developed a novel multi-agent framework specifically tailored for ophthalmic tasks, where a Decision Agent dynamically selected external tools and their invocation order to construct a traceable diagnostic reasoning workflow. By introducing cyclic tool verification, closed-loop auditing, and iterative error correction mechanisms, the framework imposed process-level constraints on medical reasoning trajectories, thereby reducing hallucinated diagnostic conclusions and improving the reliability and traceability of decision-making.

Behavior Safety Guidance. By leveraging the ability of models to adapt to new tasks and information without requiring parameter updates, In-Context Learning (ICL) holds promise for mitigating hallucinations [47]. Vu et al. [92] showed that utilizing up-to-date evidence retrieved from search engines as prompts can improve factual accuracy and reduce erroneous outputs.

Different fundamentally from only modifying input prompts, reasoning-based mitigation methods focus on structuring and verifying the model's internal reasoning, can better address complex tasks that require logical and step-by-step thinking. Zhao et al. [126] proposed a Logical Chain-of-Thought (LoT) structure, which guides large language models to reason step by step along explicit, verifiable logical paths, thereby reducing the probability of logical leaps and factual hallucinations during free-form generation. Additionally, by having models explicitly generate self-reflective rationales and confidence expressions, Xu et al. [102] reshaped model output format and reasoning habits, guiding them to incorporate reliability judgments into the generation process. This integration encouraged models to adjust or revise generated content in low-certainty scenarios as well as effectively mitigated hallucinations. Dhuliawala et al. [15] proposed a Chain-of-Verification (CoVe) framework that enabled LLMs to perform multi-step self-verification over initially generated responses, thereby improving factual accuracy and mitigating hallucinated content through iterative verification and revision. The survey [2] discovered that effective hallucination mitigation requires complementary combinations of methods that target different failure modes, rather than simply stacking redundant strategies with similar limitations. In particular, combining reasoning-based correction techniques with external evidence-augmented knowledge alignment methods achieves more stable improvements in both factuality and faithfulness.

- **Factual error.** Factual error is the semantic deviation where the agent's output conflicts with objectively verifiable facts, focusing on the assessment of correctness at the outcome level. Such erroneous information may be propagated increasingly when users trust the reliability of agent-generated content, thereby amplifying its negative impact. In medical contexts, this can further lead to the dissemination of misinformation that adversely affects individual clinical decisions and broader public health outcomes [61]. By contrast, from the perspective of generative mechanisms, hallucination tends to characterize the phenomenon where a model produces information when lacking the constraints of reliable evidence. Consequently, we discuss factual error as a distinct risk associated with foundation models that can be analyzed independently.

Behavioral Safety Constraints. To automatically detect factual errors, Cohen et al. [11] proposed a zero-shot cross-examination framework inspired by truth-seeking mechanisms in legal interrogation. The framework introduced two interacting roles: the EXAMINEE, which generated a factual claim, and the EXAMINER, which evaluated the correctness of that claim. Through prompting-based multi-turn interactions between the two language models, the method exposed semantic inconsistencies and unsupported statements, hence improving factual error detection. This cross-examination mechanism effectively introduces external verification constraints into the generation process, reducing the risk of propagating factually inconsistent outputs.

Behavior Safety Guidance. Beyond self-verification mechanisms, recent studies have explored retrieval-grounded safety guidance to mitigate factual errors in agent outputs. Li et al. [45] proposed Retrieval-Augmented Correction (RAC), which retrieved external evidence to identify and revise unsupported statements after generation without requiring additional model fine-tuning. Further extending toward memory-aware generation frameworks, Chen et al. [7] introduced Explicit Working Memory (EWE), which continuously updates an external memory module using online fact-checking feedback during long-form reasoning and generation. By dynamically revising unsupported claims throughout the reasoning trajectory, EWE reduces factual drift and improves consistency across extended agent interactions.

- **Inherent bias.** Inherent bias in agentic systems refers to systematic and unfair favoritism or prejudice, which can lead to discriminatory and stereotypical outputs. The use of unrepresentative data, the inappropriate use of AI algorithms in model development and the improper user interactions with the model are three factors responsible for inherent biases in AI models. Such flaws can be amplified and embedded into automated decision-making pipelines in real-world applications, leading to inconsistent or even discriminatory behavior against specific groups. A recruitment model may screen out qualified candidates due to gender or racial biases, while a medical diagnostic system may yield missed or erroneous diagnoses for minority populations.

Behavioral Safety Constraints. Hanna et al. [27] emphasized mitigating bias through joint interventions across the data, model and application layers, specifically by improving the representativeness and annotation fairness of training data to reduce biased inputs at the source. Hasanzadeh et al. [28] focused on healthcare scenarios, highlighting the introduction of fairness constraints and clinical knowledge-based calibration mechanisms to help ensure more consistent diagnostic and decision-making outputs across diverse patient populations. By incorporating explicit constraint mechanisms into the learning and decision-making processes, the prediction space is effectively regulated to reduce the risk of bias propagation in downstream applications. Furthermore, Xu et al. [104] proposed a multi-objective approach within a multi-agent framework (MOMA) to reduce social bias while preserving task accuracy. MOMA deployed a masking agent to identify and suppress potential bias-related features, as well as a balancing agent to introduce neutral or rephrased vocabulary during generation. This closed-loop optimization over generation and feedback progressively mitigated socially biased outputs, and demonstrated the potential of causal interventions for debiasing.

Behavior Safety Guidance. Recent studies have explored self-debiasing guidance mechanisms to suppress biased outputs. Gallegos et al. [19] proposed a zero-shot self-debiasing technique that enables large language models to recognize and iteratively reduce stereotypical generations through self-reflection prompts, without modifying model parameters or training data. By incorporating internal bias-aware feedback into the decoding process, the framework lightweight behavioral guidance for safer and fairer text generation. Beyond single-model guidance, Owens et al. [70] proposed a multi-LLM debiasing framework in which multiple language models jointly evaluate and refine generated outputs to suppress implicit negative stereotypes. This cooperative feedback mechanism improved fairness robustness and demonstrated the potential of distributed safety guidance for mitigating bias propagation in agent-based systems. Extended into multimodal agent systems, Cheng et al. [9] proposed a social debiasing guidance framework by introducing counter-stereotype guidance during multimodal alignment and generation. The approach continuously steered the model away from biased semantic associations, thereby improving fairness and neutrality in outputs while preserving downstream task performance.

- **Alignment tax.** Alignment tax in our survey refers to the intrinsic trade-off between safety alignment and agent utility, where improving safety capability often comes at the cost of reasoning degradation [37], execution impairment, and over-refusal behaviors in autonomous agents.

Behavioral Safety Constraints. Deliberative Alignment [24] explicitly retrieved and reasoned over safety specifications prior to response generation, thereby shifting safety alignment from implicit policy imitation toward interpretable deliberative reasoning. Through policy-grounded compliance assessment before action generation, the framework more effectively differentiated harmful and benign requests, thereby reducing over-refusal behaviors while maintaining benign task utility. Failure-Trajectory Evolution(FATE) framework [117] adopted an online policy optimization perspective, and employed Pareto Front Policy Optimization to select corrective samples from agent-generated failure trajectories that jointly balance safety, utility, and refusal calibration. It thereby transformed multi-objective constraints into effective signals for policy evolution, substantially improving safety while preserving the agent’s task execution performance.

Behavior Safety Guidance. The SafeConstellations [59] framework dynamically captures task-specific internal representational trajectories during the inference stage, and precisely steers requests with benign intent that would otherwise trigger safety mechanisms from a refusal pathway toward a normal execution pathway. Furthermore, AgentAlign [121] mitigated the utility degradation caused by excessive safety alignment through behavior-chain-driven alignment data synthesis. Specifically, the framework leveraged abstract behavior chains as cognitive guidance to construct benign agent trajectories while incorporating sensitive edge-case scenarios to calibrate the boundary between harmlessness and helpfulness, thereby preserving benign task executability under safety constraints.

3.1.2 Intention drift. Intention drift refers to agents’ inconsistent with initial human-assigned constraints and instructions, as multiple rounds of interaction proceeded. This unintended deviation can self-consolidate and self-reinforce through the dynamic feedback loop of autonomous interactive agents, therefore introducing potential behavioral safety risks to real world.

Based on an analysis and summary of the literature reviews, we outline the following intention deviation categories. ❶ **Goal deviation.** Arike et al. [3] found that agents exhibited tendency to gradually deviate from their originally specified objectives during prolonged interactions or under competing goals. Such goal drift became increasingly severe as interaction horizons and contextual complexity grow, revealing the limited capability of current agents to consistently preserve long-term objectives. ❷ **Semantic drift.** Yu et al. [118] observed that tool-enabled agents spontaneously developed multi-step strategies to circumvent safety constraints. Even if the outputs remain syntactically valid, the underlying decision-making logic has deviated from the initial configuration [77], demonstrating the inherent structural challenges in maintaining the consistency of execution-oriented behavior during decision-making process [76]. ❸ **Multi-agent divergence.** Han et al. [26] identified a critical post-deployment risk termed Alignment Tipping Process (ATP), in which the continual self-evolution of LLM agents gradually abandoned their initial alignment constraints under environmental feedback. In multi-agent settings, such violations could further diffuse across interacting agents [5], ultimately leading to persistent divergence from human intent and collective misalignment. Besides, role drift occurred when agents gradually deviated from their designated responsibilities and began to perform actions outside their expertise in extended collaboration [93]. For instance, a coder agent might start providing feedback, while a reviewer agent might begin writing implementation code, leading to degraded overall performance.

Behavioral Safety Constraints. The consequences of intention drift are particularly severe in long-running autonomous interactive tasks. Without proper mechanisms to detect and correct these deviations, risks compound across interaction turns, ultimately causing behavioral failure or requiring significant human intervention. Rath et al. [76] proposed a lightweight safety constraint framework RoleFix, which introduced a structured protocol requiring agents to explicitly declare role and commitments. The mechanism further incorporated a hybrid drift detector combining rule-based checks with semantic judgment, together with a reflection–reassignment–resumption cycle for automatic correction. Su et al. [82] proposed the Reflective Risk-Aware Agent Architecture (R2A2).

By incorporating risk assessment, behavioral reflection, and policy verification mechanisms into the agent decision-making pipeline, the framework enabled agents to continuously detect potentially unsafe behaviors during execution and dynamically adjust subsequent actions in accordance with predefined safety policies. R2A2 represented a paradigm shift from outcome-level filtering toward process-oriented governance, embedding safety constraints into the entire lifecycle of agent, thus mitigating the risk of behavioral loss of control in complex opening environments.

Behavior Safety Guidance. While GUI agents have shown strong performance under explicit and completion instructions, real-world deployment requires handling ambiguous instructions and inferring users' latent intents. Liu et al. [57] proposed a hierarchical implicit intent alignment framework for personalized GUI agents. By exploiting long-term user-centric records, the PersonalAlign framework aligned their actions with hierarchical implicit intents spanning user preferences and routine behavioral intents. Such memory-driven behavioral guidance enabled agents to incrementally update user's behavior and facilitate proactive assistance.

In addition, Becker et al. [5] proposed DRIFTJudge and DRIFTPolicy to identify and mitigate goal drift at test-time. While DRIFTJudge identified drifts acting as an LLM-as-a-judge, DRIFTPolicy employed a feedback agent to provide constructive suggestions to participants, guiding the agents to self-calibrate their discussion and refocus on the initial task objectives. The Residual Knowledge Stability (RKS) framework [77] introduced a systematic approach for evaluating semantic consistency and residual knowledge preservation across sequential task executions. By quantifying accumulated semantic drift in long-horizon interactions, the framework provided interpretable signals for intention drift detection and offered insights into the development of more semantically stable agent architectures. During multi-turn interactions, even minor errors or stylistic deviations of an agent can be continuously amplified through the use of its own outputs as subsequent inputs, thereby forming a negative feedback loop. Adaptive Behavior Anchoring (ABA) [76] leveraged dynamically injected baseline examples to force the system back to the initial task specifications. Consequently, it effectively interrupted the accumulation of errors and semantic deviations in the auto-regressive process, ensuring robust intention alignment over long-term interactions.

3.1.3 Memory shortcuts. Memory shortcuts reveal that agentic systems often behave as lazy learners [85], directly reusing superficially relevant past memories instead of performing semantic reasoning and decision-making. This is a procedural deficiency in foundation models when leveraging memory for reasoning. For example, upon detecting the token "Starbucks", a personal assistant agent may retrieve a prior behavioral pattern such as "purchase an iced Americano" without re-evaluating the user's current intent or context.

Such recall relies on associative correlations along historical trajectories rather than causal environmental structures [16], leading to poor generalization under out-of-distribution scenarios. On the one hand, as memory frequency and recency accumulate, agents tend to over-rely on specific interactions, which reduces their adaptability to newly emerging contexts and dynamic environments [31, 66]. On the other hand, through iterative feedback and memory updating mechanisms, these spurious correlations can become progressively self-reinforcing [84]. Actions induced by shortcut memories generate new trajectories that further strengthen the same biased associations, ultimately forming a closed feedback loop that amplifies decision bias and undermines robust reasoning.

Behavioral Safety Constraints. Tang et al. [84] designed a plug-and-play causal-aware memory calibration framework (CAMEL) to identify and mitigate shortcut-driven memory behaviors caused by spurious trajectory correlations in agentic systems. CAMEL constrained unsafe reliance on memory shortcuts, by filtering context-dependent noisy memories during the write phase, and re-ranking candidate memories according to causal invariance and cross-environment stability during retrieval. Larimar [12] designed a distributed episodic memory controller, which maintained contextual consistency and experiential controllability during long-range reasoning, thereby alleviating the model's shortcut-style dependency on locally highly salient memories.

Behavior Safety Guidance. A-MemGuard [98] compared reasoning paths derived from multiple related memories to detect anomalies. Based on this consensus-based validation, a dual-memory structure detected distilled lesson memory as reference to avoid repeated erroneous actions, presenting a proactive and experience-driven defense strategy. HIAGENT [33] introduced a trajectory retrieval module that selectively recalled complete historical action-observation pairs during long-horizon task execution. When sub-goal execution fails, the detailed trajectory information could facilitate failure attribution and corrective replanning, while previously successful trajectories provide structured experiential guidance for decision-making in novel scenarios.

3.1.4 Infinite planning loop. Autonomous agents that iteratively replan based on execution feedback may get stuck in infinite planning loops, repeatedly generating ineffective or redundant action sequences without making substantive progress toward task objectives. Such pathological planning states prevent effective corrective refinement, resulting in excessive step consumption, stalled execution trajectories, and severe degradation in long-horizon task performance [115].

Recent studies suggest that lengthy contextual histories and noisy intermediate feedback can substantially impair the agent’s planning and replanning capabilities, limiting its ability to generate corrective strategies [8]. More critically, Schepanowski et al. [78] demonstrated that agents frequently entered repetitive planning cycles or selected non-progressive actions even when explicitly instructed to avoid invalid operations, revealing intrinsic limitations of current foundation models in adaptive sequential decision-making.

Behavioral Safety Constraints. Current mitigation strategies focus on hard termination mechanisms that abruptly halt execution, sacrificing productive reasoning [40]. For instance, the LoopGuards strategy [21] provided explicit termination control by setting a maximum number of iterations, a deadline, and a time budget, thereby ensuring the convergence of the planning process.

However, purely rule-based termination mechanisms often interrupt potentially recoverable reasoning trajectories, limiting agent autonomy and task completion capability. To address this limitation, recent studies have shifted toward planning-aware mitigation strategies that proactively reduce the likelihood of agents entering infinite planning loops. Li et al. [44] proposed AOP for multi-agent systems, in which a reward model ensured that every decomposed subtask was within the agent’s capabilities to prevent it from getting stuck in planning unsolvable subtasks. Khan et al. [40] introduced the Cognitive Companion, a parallel monitoring architecture with an LLM-based Companion. Like a thinking partner who remained silent during productive work but intervened constructively when obstacles arose, the Companion preserved agent autonomy while providing targeted assistance when detected infinite planning loop.

Behavior Safety Guidance. To guide autonomous agents get rid of infinite planning loops, Zheng et al. [127] proposed KnowLoop, a closed-loop planning fault detector based on uncertainty estimation. During plan execution, it proactively identified unreliable decision steps, aborting execution and requesting human intervention when uncertainty exceeded the threshold. This human-in-the-loop guidance mechanism effectively avoided execution errors or planning deficiencies. Chen et al. [8] proposed Feedback-Aware Fine-Tuning (FAFT), which integrated both positive and negative feedback for supervised fine-tuning. The method utilized detailed feedback generated by expert systems as a chain of thought to guide model learning. During inference, an "all-success" feedback prompt induced the model to generate accurate plans for complex long-horizon planning tasks. Schepanowski et al. [78] designed a dedicated loop detection mechanism, which treated an infinite loop as a termination reason. Its specific feedback modular informed the model at which action it became trapped in the loop, and instructed it to restart from the state prior to the loop.

3.1.5 Reflective rigidity. Reflection is a post-hoc correction and self-refine process [79], in which autonomous interactive agents check, evaluate and rectify their reasoning, planning and actions. The cognitive rigidity of reflection manifesting as high confidence in previous conclusion when self-evaluating, and adherence to early flawed cognition. For instance, Weatherhead et al. [97] indicated that GPT-o4-mini offered stubborn inaccurate

feedback choice rather than delivering targeted correction in the search and identification task. his inherent rigidity hinders agents from adjusting subsequent strategies based on reviewing and summarizing its experiences, also allows errors persist in the following reflection loops.

Behavioral Safety Constraints. To alleviate cognitive rigidity during reflection, Weatherhead et al. [97] argued that reflective reasoning in contemporary agents should be grounded in executable guardrails, particularly when intrinsic metacognitive capabilities remained insufficient. Instead of relying solely on self-generated reflective narratives, executable guardrails introduced externally verifiable feedback loops that enforced constraint-aware correction and execution-level validation. This paradigm shifted reflection from a purely linguistic process toward one grounded in behavioral repair and iterative reasoning refinement.

Behavior Safety Guidance. Unlike executable guardrails that depend on external verification signals, Self-Contrast [79] improved reflection through internally diversified reasoning dynamics. Through comparative evaluation across divergent reasoning trajectories, the framework exposed reasoning discrepancies and facilitated more accurate and stable self-correction. Cognition-aware Policy Optimization (COPO) [110] further alleviated rigid reflective reasoning by introducing adaptive step-level cognitive-depth allocation, which indirectly alleviated cognitive rigidity during reflective reasoning. Lee et al. [43] proposed Spike-Triggered Adaptive Reasoning Steering (STARS), a training-free framework designed to rectify cognitive rigidity by monitoring latent dynamics. STARS employed geometric trajectory analysis to diagnose whether the agent was stubbornly adhering to a flawed yet internally consistent reasoning trajectory. The method then injected state-aware language cues to steer the agent away from entrenched cognition and back toward correct logic. Tsui et al. [91] incorporated subtle cues such as the word "Wait" to significantly enhance an agent's ability and reliability in self-correcting output errors in output.

3.2 Memory Vulnerability

During the long-term operation of autonomous interactive agents, memory systems not only serve as repositories for contextual storage and experiential retrieval, but also directly influence behavioral planning and decision-making stability. However, most existing memory mechanisms in LLM-based agents adopt static hard-coded designs tailored to specific tasks. Their memory read-write policies and retrieval logic lack dynamic adaptability, making them vulnerable such as memory contamination, memory conflicts, and memory misevolution during continuous interactions, which may ultimately induce behavioral instability.

3.2.1 Memory Contamination. During the ordinary interactions of autonomous agents, previously accumulated benign information can silently mislead unrelated tasks and negatively influence user-facing decision-making and behavior without attack-driven prompt injection, which we refer to as memory contamination. Once seemingly plausible yet unverified memories enter the context window, standard memory consolidation mechanisms may inadvertently promote short-term pollution into enduring long-term memories [52]. As the number of tasks processed by agents increases, these contaminated memories continuously accumulate and intensify, becoming a dynamically deteriorating process, eventually leading to undesirable consequences [1]. Particularly when multiple agents share a single dynamic memory pool, contaminated memory can rapidly propagate throughout the collective, leading to degraded or unexpected behavior [20]. This critical security vulnerability arises from an inherently architectural design flaw: the shared conversational session or knowledge layer lacks rigorous content source auditing and scope restrictions. For instance, a user's personal agent built on OpenClaw periodically checks email in the background while the user is away. One of the messages received is a friend's advice to use a specific algorithm for legal counsel. The next time the user asks the agent to solve that legal counsel, it chooses that specific algorithm since it has already internalized the advice into memory with user's unawareness [124].

Behavioral Safety Constraints. Yang et al. [111] implemented a write-time sanitizer: Sanitized Shared Interaction (SSI). SSI interposed on the write path of the shared-state agent, whenever an interaction was about

to be persisted, SSI rewrote it into a version that retained reusable content while removing scope-bound artifacts. If such a rewrite could not be produced safely, SSI abstained and wrote nothing. Nevertheless, sanitization was effective only when shared state was conversational, indicating that shared-state agents needed artifact-level defenses beyond text-level sanitization to prevent silent cross-user failures. The Read Filtering Gate [42] ensured that agents retrieved information only from high-confidence and timely memories by evaluating source reliability and time freshness. Simultaneously, this gating mechanism could inject identity access control (ABAC) constraints, forcing memory subgraph isolation, preventing the cross-contamination of agent personas.

Behavior Safety Guidance. Interactive Memory Sharing (INMS) [20] guided agents to activate latent logical capabilities by establishing a shared memory pool and using dynamically filtered prompt-answer pairs as context examples, effectively filtering benign but irrelevant knowledge.

3.2.2 Memory Conflict. Interactive agent is equipped with long-term memory for maintaining persistent user profiles over time. However, building memory systems reliably is challenging since dynamic user-specific information leads to the coexistence of contain earlier states, later updates, contradictory statements [125], and condition-specific preferences [88]. We refer to such sophisticated coexistent phenomena as memory conflict that burdens the memory reflection stage. Notably, unresolved conflict can accumulate and undermine the reliability of personalized memory. Enhancing the capability of identifying temporally valid, factual accurate, and contextually applicable memories becomes increasingly important to ensure the agent remains aligned with current realities and user states [34].

Behavioral Safety Constraints. Common approaches to resolving memory conflicts include DELETE operation for removal of contradictory memories [10] as well as explicitly listing constrained item in prompt template which requires agent to prioritize the most recent memory [108].

Graphiti [75] employed LLMs to identify semantic contradictions between new and old knowledge, and utilized an edge invalidation mechanism to set the expiration time of old memories to coincide with the activation time of new memories—thereby prioritizing the validity of the latest information while simultaneously maintaining historical records. Selective forgetting [34] to revise, overwrite, or remove previously stored information when faced with conflicted evidence, can also help align with goals in model editing and knowledge unlearning tasks. Exploiting quality gates based on confidence scores for conflicted memory checking and periodic expiration [17] may be feasible directions for future research.

Behavior Safety Guidance. To achieve lifelong human-agent interaction, agents need to constantly memorize perceived information and properly retrieve it for response generation. While prior studies focus on getting rid of outdated memories to improve retrieval quality, we believe in the argument that though outdated, contradictory, or conditionally inapplicable, these memories still provide rich profile information of user growth and change. THEANINE [69] discarded memory removal, and managed large-scale memories by linking them as structured timelines which represented the evolution or causality of relevant past events. Meanwhile, by retrieving and dynamically optimizing these timelines based on the current context, the framework enabled the agent to generate responses which were highly consistent with the facts of past interactions.

3.2.3 Memory Misevolution. Memory-augmented agents can learn from previous trajectories to inform better decision-making [56], yet, as the flip side of the coin, they can also learn undesirable preferences, historical habits, as well as future default rules, or degrade their risk awareness while accumulating experience into memory [103]. We refer to this unintendedly self-generated deviation as memory misevolution, which is on account of over-relying on past experiences without critical reflection, and may induce emergent unexpected misbehavior characterizing as safety alignment decay and deployment-time reward hacking. For instance, a service agent even built on state-of-the-art LLMs proactively offers refunds to all dissatisfied customers though not asked to. This overgeneralization of experience means the agent may exploit a flawed heuristic from its memory that is correlated with high historical task success [80].

Behavioral Safety Constraints. Explicit gating [120] allowed the agent to autonomously choose when to integrate memories and treated raw episodes as first-class evidence, avoiding memory faulty caused by forcibly rewriting original experiences. CAMS [14] utilized a five-layer zero-trust architecture integrating semantic intent interception, temporal drift monitoring, and graph relationship reconstruction to identify and intercept progressive memory misevolution in real-time. Augmented by the WriteGuard write-protection mechanism and LTMScanner periodic retroactive auditing, the framework effectively captured error evolution sequences that evaded detection. Consolidation channel introduced regression guards [101] could prevent the misevolutionary memory from becoming permanent parameterized rules. Furthermore, it allowed for timely rollback upon detecting anomalies, restoring the system to a safe parameterized state and preventing the deep internalization of errors.

Behavior Safety Guidance. Carefully curated meta-prompts can mitigate harmful behaviors introduced by memory misevolution. Specifically, the approach involves instructing the agent to treat retrieved memories merely as references [80]. However, prompt-oriented methods address the symptoms, not the root cause. Future works are expected to focus on improving the memory retrieval mechanism itself. Furthermore, a task-specialized memory harness framework enables to abstract the agent memory mechanism into a programmable memory system defined through Python programs, where memory read-write interfaces, storage structures, and retrieval logic are encapsulated within a unified architecture. MStar [71] introduced a reflective code evolution mechanism, using a large language model as a reflector to analyze failure cases in the evaluation and optimize the prompt that guided the agent's behavior.

3.3 Tool-invocation Vulnerability

With the rapid advancement of large language models, autonomous agents can perform complex tasks in open-ended environments via invoking external tools and interacting with real-world systems [62]. The autonomy of the agent is largely determined by the accessible tools, the broader tool access implicates greater capacity to plan, reason, and act by invoking tools, as well as yields broader tool-driven agency [6]. However, this expanded capability amplifies security risks, causing agents to collapse or behave abnormally. From the perspective of tool invocation procedure, our discussed agent's internal tool-use vulnerabilities are fourfold: reading tool documentation, selecting tools, invoking tools, and handling tool responses.

Existing agent architectures typically assume that the tool registry is trusted, and the agent automatically selects tools based on their names or descriptions. This allows attackers to deceive agents by forging tools with the same or similar names and inducing agents to call malicious interfaces [65].

When selecting tools, agents commonly face risks from an unbalanced tool-driven agency due to the lack of self-awareness, which manifests in two forms. Excessive agency arises when agents are provisioned with unnecessary or overly permissive tools, even when internal reasoning suffices, resulting in stagnant internal capability despite correctness [94]. Conversely, insufficient agency arises when agents lack or misidentify required tools, preventing correct task execution [6]. Additionally, Xiong et al. [100] revealed that increasing the model size does not significantly improve the reasoning of tool invocation and may make agents more vulnerable to attacks resembling normal user instructions.

When executing tools, agents often operate in a step-by-step manner, lacking the capability to holistically assess the cumulative effects produced by the entire action sequence [46]. The aforementioned imbalance will lead to tool overuse in agent systems, where models unnecessarily rely on external tools for tasks solvable with parametric knowledge, increasing computational overhead [74]. This not only leads to unnecessary resource consumption but can also confuse the model, ultimately degrading performance.

Many agent frameworks directly write tool-returned content into the context or memory [114], allowing attackers to inject prompt injections, unauthorized instructions, or fabricated states into tool outputs, thereby manipulating subsequent decision-making.

Behavioral Safety Constraints. Monitoring step-level tool invocation behaviors in real time and proactively intervening before unsafe execution is critical for agent deployment. Zero-trust registration mechanism [65] verified and isolated the identity, permissions, source, and invocation strategy of tools. ToolFuzz [60] automated testing of tool documentations to detect errors in advance. From the perspective of epistemic necessity, ToA [94] reconceived agents as tool-use decision makers, insisting that external tool-interaction was justified if and only if the remaining uncertainty required to complete a task could not be resolved through internal reasoning alone. SMART [74] enhanced an agent’s self-awareness through a strategic model-aware reasoning paradigm to optimize task handling and reduce tool overuse. By aligning subjective perception with knowledge boundary, SMARTAgent was able to make more intelligent decisions on where to rely on external tools or internal parametric knowledge. TS-Guard [62] was a guardrail model for step-level tool invocation safety detection that was trained via multi-task reinforcement learning tailored for agent security. The model proactively detected unsafe tool invocation actions before execution by reasoning over the interaction history, and provided interpretable and generalizable safety analysis. Then, TS-Flow delivered preexecution feedback to the agent for dynamically correcting reasoning and enhancing tool invocation safety.

Besides, without altering an agent’s internal reasoning, AgenTRIM [6] addressed tool-driven agency risks through complementary offline and online phases. An offline tool extractor built a verified tool inventory prior to deployment. At runtime, a tool orchestrator enforced least-privileged tool access per step through adaptive filtering and status tracking, so that agents retained only the tools required at each inference step and ensure safer tool use. HDPO [107] addressed the high latency and inference interference caused by the blindly calling tools of multimodal agents. It utilized conditional advantage estimation to optimize tool-calling simplification while maintaining task accuracy, thus resolving the gradient interference problem in reinforcement learning and enabling more rational tool invocation.

Behavior Safety Guidance. Current multi-agent systems often face inherent failures in actual deployment, especially in SME automation, where agents fail to invoke tools correctly or fail to execute, stemming from unreliable tool invocation decisions. Huang et al. [35] proposed a comprehensive diagnostic framework that leveraged big data analytics to quantitatively evaluate procedural reliability of agents in the tool invocation process. It aimed to help developers systematically identify and locate the root causes of internal errors such as tool invocation failures, process interruptions, or model phantoms. VIGIL [49] introduced a verify-before-commit mechanism that performed safety verification, semantic consistency checking, and policy filtering on tool outputs before they entered the agent’s memory or execution chain, thereby addressing the lack of trustworthy boundaries for tool-generated content.

3.4 Multi-agent Architecture Vulnerability

Multi-agent systems extend single-agent capabilities to intricate task scenarios by decomposing and delegating tasks among agents with distinct roles or identities [30]. Agents can share knowledge, engage in interactive discussions, execute subtasks, and align their efforts toward shared goals [90]. This unleashes collective intelligence and enhances capabilities across domains ranging from strictly rational tasks to highly exploratory ones. However, multi-agent coordination mechanisms, state sharing, authority delegation, and distributed tool access can introduce emergent risks due to inherent agent incapability or vulnerability [68].

Drawing on the two dominant interaction paradigms [25]—cooperation and competition—we discuss the corresponding key failure modes: miscoordination and over-competition. Miscoordination occurs when multiple agents share aligned goals yet fail during collaborative task execution. In cooperative settings, agents typically

treat each other's messages as trustworthy. This blind mutual trust amplifies potential risks of over-exposure and over-authorization [105]. Asynchronous state updates or communication delays cause agents to make decisions based on outdated states, resulting in de facto miscoordination. Because information flows in multi-agent systems are deeply interconnected and system states depend heavily on prior execution rounds, failures such as hallucinations in a single agent can propagate and be amplified along the collaboration chain [87]. Small deviations can thus cascade into severe consequences and ultimately degrade overall system performance. In Generative Multi-Agent Systems (GMAS) for the telecommunications domain, even the diversity of agent roles can become a latent source of cooperative risk [67]. When agents designed with different role profiles—such as coding style or planning orientation—collaborate under mismatched knowledge backgrounds, they are prone to generating conflicting actions or interpretative discrepancies, leading to system-level instability. Moreover, as the number of collaborating agents increases, the complexity of task allocation and conflict management grows exponentially, as well as consensus mechanisms undergoes structural degradation—specifically, coordination drift [39]. This phenomenon manifests as intensified inter-agent conflicts, task redundancy, coordination failures, and imbalances in allocation preferences, reflecting a systemic breakdown in the internal collaborative logic of the system as it evolves over time within complex environments [76].

Unlike the miscoordination that arises under cooperation, the competition paradigm leads to a different failure mode: over-competition. Embedding competitive mechanisms into multi-agent systems can achieve optimal agent-task matching and resource allocation in large-scale, complex scenarios [53, 123]. Yet such mechanisms can also induce over-competition. When a system simultaneously demands internal cooperation and external competition—such as in multi-agent team contests—agents may sacrifice team interests for individual ranking and engage in excessive rivalry. In multi-agent debates, prompts invoking survival instincts such as “failure leads to elimination” cause agents to exhibit hyper-competitive behaviors under competitive pressure, thereby degrading task performance. They shift from discussing the overall task to narrowly fixating on specific points, emphasizing competition over task resolution. They also display higher frequencies of boasting, inflammatory rhetoric, and an aggressive tone [58]. These negative effects become more pronounced in subjective tasks, where no objective ground truth exists. Beyond behavioral deterioration, in competitive scenarios the game dynamics among agents create an unstable environment for one another, potentially driving the system into unsafe or inefficient Nash equilibria.

Behavioral Safety Constraints. M2I2 [83] introduced two self-supervised auxiliary tasks—masked state modeling and joint action prediction—enabling the agent to reconstruct the global state from limited information and implicitly infer teammates' intentions, thereby enhancing the accuracy of collaborative decision-making and maintaining alignment with overall goals and strategies. The AgentChain framework leveraged blockchain technology to transform the centralized control inherent in traditional collaboration frameworks into a distributed semantic consensus process based on peer-to-peer collaboration, better ensuring the semantic consistency of the generated answers.

Sensitive-Information Repartitioning [105] sharded sensitive information and employed cross-validation across multiple agents to ensure that no single pathway can access complete confidential data. The Guardian-Agent provided online supervision and compliance audits. Prior to an agent delivering its final response, real-time checks and interceptions were executed based on preset policies—such as the Principle of Minimum Necessary Information (MNI)—thereby rectifying improper conduct before any data leakage occurred.

Introducing a fair judge as an external agent could significantly mitigate excessive competition which provided objective and task-centric feedback. In the absence of such external feedback, peer-as-court mechanism, where a collective decision was made by majority voting to select the worst-performing solution, could also alleviate over-competition.

AgentNet [112] adopted a dynamically evolving graph topology, where both nodes (agents) and edges (agent-to-agent connections) were adjusted in real time based on task demands and agent performance. The system

optimized information flow and task allocation through a directed acyclic graph (DAG) that supported flexible collaboration and prevents circular dependencies, reducing the exposure of sensitive information.

Drift-Aware Routing (DAR) [76] enforces behavioral safety constraints at the architectural level by integrating Agent Stability Index (ASI) scores into task assignment decisions. This mechanism prioritizes the authorization of high-stability agents and enforces contextual resets on agents that trigger the drift threshold. By rigidly cutting off unreliable states to reconstruct the execution environment, it ensures boundary safety and collaborative consistency in multi-agent systems over long-term interactions.

Behavior Safety Guidance. To further improve the resilience of multi-agent systems, Huang et al. [36] introduced an additional Inspector agent to intercept all messages propagated among agents, check for errors, and correct them. To address cascading failures caused by multi-tier architectures, Nguyen et al. [68] drawn attention to the necessity of leveraging distributed trace correlation to trace back to upstream fault sources. Drawing on ProvAgent’s approach to detecting the sources of Advanced Persistent Threats [109], future research may explore graph contrastive learning. This could enable fine-grained identity–behavior consistency across multiple agents, thereby facilitating the attribution of errors to specific parties. Additionally, research on content verification and automatic rollback techniques holds considerable promise for the future, as these techniques may help agents restore the last known good state after an erroneous agent has been traced, thereby enabling task completion to resume.

4 External Loss-of-control Risk

In this section, we focus on the external loss-of-control risks faced by autonomous interactive agents during training and runtime. Malicious attacks from external sources can amplify the endogenous vulnerabilities mentioned above, leading to more harmful behavioral consequences. Based on the source of the attack, we categorize these external loss-of-control risks into four types: supply-chain manipulation, user-level manipulation, environment-level manipulation, and cross-agent interaction risks. Among these, supply-chain manipulation is training-phase risk, whereas the other three types are runtime risks. We first introduce the risks posed by each type. Then we elaborate on the corresponding attack research. Finally, we summarize both existing and future behavioral safety defenses.

4.1 Supply-Chain Manipulation

Supply-chain manipulation serves as the entry point exploited by attackers. We outline it from two perspectives: data poisoning and backdoor attacks.

4.1.1 Data Poisoning. Data poisoning attacks inject malicious samples during pre-training, fine-tuning, or reinforcement learning from human feedback (RLHF) to induce models to internalize harmful behavioral patterns [86], thereby eliciting targeted or globally harmful outputs during inference and degrading overall agent performance [18, 63, 96]. Unlike the backdoor attacks discussed later, data poisoning is a training-time attack that does not rely on a specific trigger [29].

Current data poisoning in the agent domain primarily focuses on contaminating factual knowledge, rewards, preferences, and trajectories. **Fact poisoning** alters specific factual knowledge while preserving overall model utility. The poison pills strategy [116] introduced minimal yet critical inaccuracies into originally true knowledge (e.g., altering details such as dates, names, or locations). Fine-tuning open-source models on such toxic data significantly degrades model performance. **Reward poisoning** can occur by manipulating the agent’s rewards to mislead the victim’s training process [89] in reinforcement learning agents. During training, the adversary executes its poisoning strategy while the agent refines its policy over time under this adversarial setup. This manipulated learning process increases the likelihood of specific undesirable harmful events [22]. **Preference poisoning.** Given the reliance of reward models on pairwise preference learning, Wu et al. [99] proposed a

gradient-based framework and a class of Rank-by-Distance (RBD) greedy heuristics. An attacker can flip a small fraction of preference labels to degrade target outcomes, causing substantial downstream impact on policies learned via RLHF. **Trajectory poisoning.** Hsu et al. [32] introduced Trap, a heuristic poisoning attack that leveraged a prefix-suffix method to efficiently generate adversarial pseudo-trajectories, substantially increasing the occurrence of attacker-specified target patterns.

Behavioral Safety Defense. Defending LLM-based agents against data poisoning attacks primarily centers on data and model integrity, filtering poisoned examples, and robust training [54]. Zhu et al. [129] proposed two provable and practical watermarking methods that achieved high-probability identification of toxic data under watermark length constraints, thereby ensuring transparency of poisoned datasets. Inspired by this, designing watermarking techniques with cryptographic signatures to verify data provenance and blockchain-enabled audit trails offers a promising direction for ensuring data auditing and accountability in agent ecosystems. Moreover, identifying anomalous poisoned samples and removing them from the training dataset can effectively mitigate poisoning attacks. Tashman et al. [89] proposed a lightweight real-time defense based on statistical filtering and reward clipping to mitigate reward poisoning attacks against DRL agents in RIS-enhanced CRNs, where mean and standard deviation dynamically adapted to gradual environmental changes while filtering out anomalous deviations. Additionally, applying differential privacy to the dataset or employing adversarial training cycles can enhance model robustness against poisoning attacks.

4.1.2 *Backdoor Attacks.* **Behavioral Safety Constraints. Behavior Safety Guidance.**

4.2 User-Level Manipulation

4.2.1 *Direct Prompt Injection.* **Behavioral Safety Constraints. Behavior Safety Guidance.**

4.2.2 *Jailbreak.* **Behavioral Safety Constraints. Behavior Safety Guidance.**

4.2.3 *Long-Horizon Manipulation.* **Behavioral Safety Constraints. Behavior Safety Guidance.**

Detection-based guardrail such as PIGuard, DataSentinel and Attention Tracker train external classifiers to flag suspicious segments.

4.3 Environment-Level Manipulation

4.3.1 *Indirect Prompt Injection.* **Behavioral Safety Constraints. Behavior Safety Guidance.**

4.3.2 *Retrieval Poisoning.* **Behavioral Safety Constraints. Behavior Safety Guidance.**

4.3.3 *Memory Poisoning.* **Behavioral Safety Constraints.**

Behavior Safety Guidance.

Damage to long-term memory can stealthily redirect agent behavior, resulting in attacks that are more persistent and covert than those enabled by existing techniques. Srivastava et al. propose a quantitative metric, the **Poisoned Retrieval Proportion (PRP)** [81], to measure the bias induced by memory poisoning. Validation is conducted on the DataInterpreter agent within MetaGPT. Experimental results demonstrate that even when poisoned records constitute only a minimal fraction of the entire memory repository, the PRP value can reach as high as 47.9 %, indicating that contaminated memory can be activated with high probability during normal task execution.

Memory Control Flow Attack (MCFA) [106] refers to a novel class of attack in which an adversary contaminates the long-term memory of an intelligent agent, thereby manipulating the sequence of tool calls, decision-making pathways, and behavioral workflows, causing the agent to persistently deviate from its original objectives in subsequent tasks. This attack transcends the transient effects of conventional prompt injection, revealing the enduring capacity of memory to dominate the control flow of agent behaviors.

Behavioral Safety Constraints.

Role-Based Memory Segregation (RBMS) [106] enforces behavioral constraints by establishing dual channels for system rules and user preferences, coupled with a strict priority hierarchy. This mechanism compels the agent to adhere to immutable rules preferentially in conflict scenarios, thereby safeguarding control flow integrity through architectural segregation and logical separation of authority, while defending against potential memory poisoning risks.

4.4 Cross-Agent Interaction Risks**4.4.1 Collusion.****4.4.2 Worm Propagation .****5 Community Safety****5.1 Protocol and Interface Risks****5.1.1 Communication Protocol.****5.1.2 APIs.****5.2 Third-Party Extension Risks****5.2.1 Plugins .****5.2.2 Tools.****5.3 Skill Risks**

HarmfulSkillBench [38] establishes a hierarchical risk taxonomy comprising 21 categories of harmful skills. Tier 1 includes 14 categories of prohibited capabilities, such as cyberattacks, fraud, and weapon development, with an emphasis on directly blocking malicious functionalities. Tier 2 covers seven categories of high-risk professional skills, including medical, legal, and financial services, which require human supervision and explicit AI identity disclosure to mitigate behavioral safety risks arising from the misuse of externalized agent capabilities.

Skill-reading Exploit is an attack paradigm targeting open agent skill ecosystems, in which a malicious skill illegitimately accesses the description metadata, execution context, and internal capability states of other skills to conduct cross-skill information theft and capability abuse. This attack exposes critical security weaknesses in agent communities, including insufficient skill isolation, ineffective permission boundaries, and the lack of security governance for third-party skill supply chains [38].

Behavioral Safety Constraints.**Behavior Safety Guidance.**

Harness Engineering. Rather than treating LLM agents merely as parameterized reasoning systems, recent studies increasingly interpret agent capability as emerging from the interaction between foundation models and external runtime infrastructures. Within this trend, Harness engineering has gradually become a central paradigm for improving the reliability, controllability, observability and governability of agent systems. Zhou et al.[128] frames this transition through the lens of externalization, arguing that many cognitive burdens previously handled implicitly within model parameters are progressively relocated into explicit external structures. Drawn on Donald Norman’s theory of cognitive artifacts, this perspective emphasizes that the role of external infrastructure is not simply to augment model capability through auxiliary modules, but to transform task representations into forms that can be processed more reliably by the model.

Under this framework, Harness Engineering is generally regarded as a unified orchestration layer that coordinates multiple externalized components within a runtime environment. In particular, Memory systems externalize state persistence and long-term continuity, Skills externalize procedural expertise and reusable operational knowledge, and Protocols externalize interaction structures and coordination rules among agents, users, and tools. The harness subsequently integrates these components through mechanisms such as control-flow management, sandboxing, permission policies, human oversight, structured feedback, and context-budget regulation, thereby constraining and stabilizing agent behavior during execution. From this perspective, current agent development can be interpreted as an evolutionary transition from weights to context and ultimately to harness, reflecting a broader shift from model-centric intelligence toward distributed cognitive infrastructure. Nevertheless, existing research remains insufficient in addressing the additional complexity introduced by harness-based systems, particularly with respect to security risks, governance overhead, and the absence of unified evaluation methodologies.

Stemming from the architectural separation between security mechanisms and the agent's control execution lifecycle, existing agent frameworks commonly suffer from structural deficiencies such as insufficient context awareness, inter-layer isolation, and a lack of resilient coordination mechanisms. Lin et al. [50] propose a hierarchical, dynamic security architecture with adjustable risk levels that embeds defensive mechanisms throughout the entire lifecycle of agent tool execution, thereby enabling a unified and coordinated system-level response to compound attacks. The framework establishes a four-tier defense system corresponding to four critical stages of agent execution: context filtering during input processing; causal validation during decision-making; tool control via privilege separation during action execution; and adaptive rollback with graceful degradation during state updates. Since information is exchanged between the various defense layers via explicit feedback pathways, detections made at one stage can propagate backward to reinforce the protective strategies of another stage, thereby establishing a cross-lifecycle, collaborative defense mechanism. This design ensures that security capabilities no longer function merely as external, add-on filters, but are instead intrinsically embedded within the complete execution lifecycle of the intelligent agent.

6 System Safety

6.1 Resource Exhaustion

6.2 Isolation and Permission Failures

6.3 Runtime Execution Vulnerabilities

Behavioral Safety Constraints.

AGENTWARD [122] proposes a lifecycle security architecture designed for autonomous agents, extending the scope of security boundaries from content security to the realm of system security. By embedding layered security control mechanisms across the stages of initialization, input processing, cognition, memory, decision-making, and execution, it establishes system-level constraints on the agent's capability invocation, resource access, and execution behaviors.

Behavior Safety Guidance.

7 Evaluation

7.1 Intrinsic Security Risks

AndroidIntent [57] is a novel benchmark designed to evaluate a GUI agent's ability in resolving vague instructions and providing proactive suggestions by reasoning over long-term user records. The benchmark utilizes a hierarchical filtering-verifying strategy to ensure high-quality annotation, filling a critical gap in the availability of long-term, user-centric evaluation tools for personalized agents.

AgentMisalignment [64] is a specialized benchmark designed to evaluate the propensity of LLM-based agents to pursue unintended or misaligned objectives in realistic scenarios. The benchmark covers multiple representative forms of unsafe agent behaviors, covering oversight evasion, shutdown resistance, sandbagging, and power-seeking tendencies.

DRIFT-BENCH [4] is the first diagnostic benchmark designed to evaluate the pragmatic capabilities of LLM agents in resolving systematic cooperative breakdowns through multi-turn clarification across both state-oriented and service-oriented environments.

SAD. As large language model-driven agents gradually acquire the capabilities of autonomous planning, tool invocation, and environmental interaction, researchers have begun to realize that relying solely on traditional task performance metrics is insufficient to fully assess the true autonomous capabilities of these agents. Compared to simple reasoning and generation capabilities, an agent capable of long-term autonomous operation also needs continuous awareness of its own state, operating environment, and capability boundaries—the so-called situational awareness. For autonomous interactive agents, a lack of accurate situational awareness can lead to problems such as incorrect tool invocation, permission overreach, environmental misjudgment, and capability hallucination, further impacting the security and reliability of the agent’s behavior.

Against this backdrop, Rudolf Laine et al. [41] systematically models the self-situational awareness capability of large language models as a quantifiable and measurable problem for the first time, and proposes a standardized evaluation benchmark for this capability, SAD (Situational Awareness Dataset). Its design pillars consist of three dimensions: self-knowledge, inferences, and actions. In terms of specific task architecture, SAD systematically examines, through seven task categories—Facts, Influence, Introspection, Stages, Self-Recognition, ID-Leverage, and Anti-Imitation—whether the model can recognize its own attributes, understand its causal influence, access internal representations, distinguish between training and deployment phases, recognize its own generated text, and execute specific instructions based on self-awareness. This design aims to go beyond simple text imitation, instead probing whether the model can translate its abstract understanding of LLM identity into autonomous planning and decision-making behavior under complex constraints.

Unlike traditional benchmarks that focus more on the correctness of task output, SAD emphasizes whether the model can form a consistent and reliable runtime self-model. This work promotes the expansion of the large language model evaluation paradigm from external task performance to internal state cognition.

HINTBench. Addressing the insufficient assessment of intrinsic safety risks faced by long-horizon agents under benign, non-adversarial conditions, we propose a framework for evaluating intrinsic risks specifically tailored to long-term autonomous execution processes. Existing research on agent safety assessment primarily focuses on externally induced risks—such as prompt injection attacks, malicious tool feedback, and environmental manipulation—while research concerning non-attack intrinsic risks remains relatively limited. This category of risk refers to scenarios where, even when user instructions are inherently benign, tool interfaces are valid, and environmental feedback is free from adversarial interference, an agent may still generate high-consequence safety risks during long-term task execution due to factors such as internal reasoning flaws, state maintenance errors, or deviations in execution strategy.

In long-horizon tasks, these intrinsic failures typically manifest with characteristics of delayed emergence and gradual accumulation. Some initial errors do not surface immediately; instead, they continuously propagate and amplify across subsequent rounds of decision-making and state updates, eventually evolving into high-risk consequences within the real-world environment—such as privacy breaches, unauthorized transactions, or the mishandling of critical resources. Consequently, relying solely on traditional attack-driven safety evaluation frameworks makes it difficult to comprehensively characterize the long-term behavioral reliability of autonomous, interactive agents in real-world deployment environments.

To systematically assess these risks, Wang et al.[95] introduce a new evaluation paradigm termed Non-attack Intrinsic Risk Auditing. Centered around the agent’s execution trajectory, this paradigm comprises three progressive auditing tasks: Trajectory-level Risk Detection, Risk-step Localization, and Intrinsic Failure-type Identification. Based on this foundation, we have constructed the HINTBench benchmark dataset. This dataset comprises a total of 629 agent execution trajectories—including 523 risky trajectories and 106 safe trajectories—with an average trajectory length of 33 steps. In terms of interaction depth and long-range dependency characteristics, HINTBench significantly outperforms existing benchmarks such as R-Judge and ATBench. Furthermore, this work introduces the Five-Constraint Taxonomy, which categorizes intrinsic failure systems—based on the core conditions an agent must satisfy to correctly execute a task—into five types: Goal Constraint, Factual Constraint, Capability Constraint, Procedural Constraint, and State Constraint. This taxonomy provides a unified theoretical framework for the fine-grained fault analysis and intrinsic safety evaluation of long-duration agent behaviors.

7.2 External Security Risks

HarmfulSkillBench [38] maps traditional software supply chain security issues into the agent skill ecosystem and systematically evaluates the behavioral security risks that emerge when agents integrate external skill repositories. By constructing diverse categories of harmful skills and adversarial scenarios, the framework analyzes external security threats arising during skill sharing, cross-skill interaction, and external capability integration, including information leakage, unauthorized invocation, and capability abuse.

8 Directions of Future Work

Acknowledgments

.....

References

- [1] Ahmad Al-Tawaha, Shangding Gu, Peizhi Niu, Ruoxi Jia, and Ming Jin. 2026. Remembering More, Risking More: Longitudinal Safety Risks in Memory-Equipped LLM Agents. *arXiv preprint arXiv:2605.17830* (2026).
- [2] Aisha Alansari and Hamzah Luqman. 2026. Large language models hallucination: A comprehensive survey. *Computer Science Review* 61 (2026), 100970.
- [3] Rauno Arike, Elizabeth Donoway, Henning Bartsch, and Marius Hobbhahn. 2025. Evaluating Goal Drift in Language Model Agents. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, Vol. 8. 192–203.
- [4] Han Bao, Zheyuan Zhang, Pengcheng Jing, Zhengqing Yuan, Kaiwen Shi, and Yanfang Ye. 2026. Drift-Bench: Diagnosing Cooperative Breakdowns in LLM Agents under Input Faults via Multi-Turn Interaction. *arXiv preprint arXiv:2602.02455* (2026).
- [5] Jonas Becker, Lars Benedikt Kaesberg, Andreas Stephan, Jan Philip Wahle, Terry Ruas, and Bela Gipp. 2026. Stay focused: Problem drift in multi-agent debate. In *Findings of the Association for Computational Linguistics: EACL 2026*. 5068–5102.
- [6] Roy Betser, Shamik Bose, Amit Giloni, Chiara Picardi, Sindhu Padakandla, and Roman Vainshtein. 2026. AgentTRIM: Tool Risk Mitigation for Agentic AI. *arXiv preprint arXiv:2601.12449* (2026).
- [7] Mingda Chen, Yang Li, Karthik Padthe, Rulin Shao, Alicia Yi Sun, Luke Zettlemoyer, Gargi Ghosh, and Wen-tau Yih. 2025. Improving factuality with explicit working memory. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 11199–11213.
- [8] Yanan Chen, Ali Pesaraghader, Tanmana Sadhu, and Dong Hoon Yi. 2024. Can We Rely on LLM Agents to Draft Long-Horizon Plans? Let’s Take TravelPlanner as an Example. *arXiv preprint arXiv:2408.06318* (2024).
- [9] Harry Cheng, Yangyang Guo, Qingpei Guo, Ming Yang, Tian Gan, Weili Guan, and Liqiang Nie. 2025. Social debiasing for fair multi-modal llms. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 1740–1750.
- [10] Prateek Chhikara, Dev Khant, Saket Aryan, Taranjeet Singh, and Deshraj Yadav. 2025. Mem0: Building production-ready ai agents with scalable long-term memory. *arXiv preprint arXiv:2504.19413* (2025).
- [11] Roi Cohen, May Hamri, Mor Geva, and Amir Globerson. 2023. Lm vs lm: Detecting factual errors via cross examination. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. 12621–12640.
- [12] Payel Das, Subhajit Chaudhury, Elliot Nelson, Igor Melnyk, Sarath Swaminathan, Sihui Dai, Aurélie Lozano, Georgios Kollias, Vijil Chenthamarakshan, Soham Dan, et al. 2024. Larimar: Large language models with episodic memory control. *arXiv preprint arXiv:2403.11901*

- (2024).
- [13] Zehang Deng, Yongjian Guo, Changzhou Han, Wanlun Ma, Junwu Xiong, Sheng Wen, and Yang Xiang. 2025. Ai agents under threat: A survey of key security challenges and future pathways. *Comput. Surveys* 57, 7 (2025), 1–36.
 - [14] T Dhivyaree, S Saravanan, et al. 2026. Cognitive Autonomous Memory Security (CAMS) against injection and extraction attacks in long-term memory of AI agents. *Egyptian Informatics Journal* 34 (2026), 100983.
 - [15] Shehzaad Dhuliawala, Mojtaba Komeili, Jing Xu, Roberta Raileanu, Xian Li, Asli Celikyilmaz, and Jason Weston. 2024. Chain-of-verification reduces hallucination in large language models. In *Findings of the association for computational linguistics: ACL 2024*. 3563–3578.
 - [16] Mengnan Du, Fengxiang He, Na Zou, Dacheng Tao, and Xia Hu. 2023. Shortcut learning of large language models in natural language understanding. *Commun. ACM* 67, 1 (2023), 110–120.
 - [17] Pengfei Du. 2026. Memory for autonomous llm agents: Mechanisms, evaluation, and emerging frontiers. *arXiv preprint arXiv:2603.07670* (2026).
 - [18] Tingchen Fu, Mrinank Sharma, Philip Torr, Shay B Cohen, David Krueger, and Fazl Barez. 2024. Poisonbench: Assessing large language model vulnerability to data poisoning. *arXiv preprint arXiv:2410.08811* (2024).
 - [19] Isabel O Gallegos, Ryan Aponte, Ryan A Rossi, Joe Barrow, Mehrab Tanjim, Tong Yu, Hanieh Deilamsalehy, Ruiyi Zhang, Sungchul Kim, Franck Dernoncourt, et al. 2025. Self-debiasing large language models: Zero-shot recognition and reduction of stereotypes. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 2: Short Papers)*. 873–888.
 - [20] Hang Gao and Yongfeng Zhang. 2024. Memory sharing for large language model based agents. *arXiv preprint arXiv:2404.09982* (2024).
 - [21] Longling Geng and Edward Y Chang. 2025. ALAS: Transactional and Dynamic Multi-Agent LLM Planning. *arXiv preprint arXiv:2511.03094* (2025).
 - [22] Omran Shahbazi Gholiabad and Mohammad Mahmoodi. [n. d.]. Targeted Poisoning of Reinforcement Learning Agents. In *Eighteenth European Workshop on Reinforcement Learning*.
 - [23] Diego Gosmar and Deborah A Dahl. 2025. Hallucination mitigation using agentic ai natural language-based frameworks. *arXiv preprint arXiv:2501.13946* (2025).
 - [24] Melody Y Guan, Manas Joglekar, Eric Wallace, Saachi Jain, Boaz Barak, Alec Helyar, Rachel Dias, Andrea Vallone, Hongyu Ren, Jason Wei, et al. 2024. Deliberative alignment: Reasoning enables safer language models. *arXiv preprint arXiv:2412.16339* (2024).
 - [25] Lewis Hammond, Alan Chan, Jesse Clifton, Jason Hoelscher-Obermaier, Akbir Khan, Euan McLean, Chandler Smith, Wolfram Barfuss, Jakob Foerster, Tomáš Gavenčiak, et al. 2025. Multi-agent risks from advanced ai. *arXiv preprint arXiv:2502.14143* (2025).
 - [26] Siwei Han, Kaiwen Xiong, Jiaqi Liu, Xinyu Ye, Yaofeng Su, Wenbo Duan, Xinyuan Liu, Cihang Xie, Mohit Bansal, Mingyu Ding, et al. 2025. Alignment Tipping Process: How Self-Evolution Pushes LLM Agents Off the Rails. *arXiv preprint arXiv:2510.04860* (2025).
 - [27] Matthew G Hanna, Liron Pantanowitz, Brian Jackson, Octavia Palmer, Shyam Visweswaran, Joshua Pantanowitz, Mustafa Deebajah, and Hooman H Rashidi. 2025. Ethical and bias considerations in artificial intelligence/machine learning. *Modern Pathology* 38, 3 (2025), 100686.
 - [28] Fereshteh Hasanzadeh, Colin B Josephson, Gabriella Waters, Demilade Adedinsewo, Zahra Azizi, and James A White. 2025. Bias recognition and mitigation strategies in artificial intelligence healthcare applications. *NPJ Digital Medicine* 8, 1 (2025), 154.
 - [29] Feng He, Tianqing Zhu, Dayong Ye, Bo Liu, Wanlei Zhou, and Philip S Yu. 2025. The emerged security and privacy of llm agent: A survey with case studies. *Comput. Surveys* 58, 6 (2025), 1–36.
 - [30] Xu He, Di Wu, Yan Zhai, and Kun Sun. 2025. Sentinelagent: Graph-based anomaly detection in multi-agent systems. *arXiv preprint arXiv:2505.24201* (2025).
 - [31] Yuki Hou, Haruki Tamoto, and Homei Miyashita. 2024. "my agent understands me better": Integrating dynamic human-like memory recall and consolidation in llm-based agents. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–7.
 - [32] I Hsu, Chih-Hsun Lin, Chia-Mu Yu, Sy-Yen Kuo, Chun-Ying Huang, et al. 2025. Poisoning Attacks to Local Differential Privacy Protocols for Trajectory Data. *arXiv preprint arXiv:2503.07483* (2025).
 - [33] Mengkang Hu, Tianxing Chen, Qiguang Chen, Yao Mu, Wenqi Shao, and Ping Luo. 2025. Hiagent: Hierarchical working memory management for solving long-horizon agent tasks with large language model. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 32779–32798.
 - [34] Yuanzhe Hu, Yu Wang, and Julian McAuley. 2025. Evaluating memory in llm agents via incremental multi-turn interactions. *arXiv preprint arXiv:2507.05257* (2025).
 - [35] Donghao Huang, Gauri Malwe, and Zhaoxia Wang. 2026. When Agents Fail to Act: A Diagnostic Framework for Tool Invocation Reliability in Multi-Agent LLM Systems. *arXiv preprint arXiv:2601.16280* (2026).
 - [36] Jen-tse Huang, Jiaxu Zhou, Tailin Jin, Xuhui Zhou, Zixi Chen, Wenxuan Wang, Youliang Yuan, Michael R Lyu, and Maarten Sap. 2024. On the resilience of llm-based multi-agent collaboration with faulty agents. *arXiv preprint arXiv:2408.00989* (2024).
 - [37] Tiansheng Huang, Sihao Hu, Fatih Ilhan, Selim Furkan Tekin, Zachary Yahn, Yichang Xu, and Ling Liu. 2025. Safety tax: Safety alignment makes your large reasoning models less reasonable. *arXiv preprint arXiv:2503.00555* (2025).

- [38] Yukun Jiang, Yage Zhang, Michael Backes, Xinyue Shen, and Yang Zhang. 2026. HarmfulSkillBench: How Do Harmful Skills Weaponize Your Agents? *arXiv preprint arXiv:2604.15415* (2026).
- [39] Weiqiang Jin, Hongyang Du, Biao Zhao, Xingwu Tian, Bohang Shi, and Guang Yang. 2025. A comprehensive survey on multi-agent cooperative decision-making: Scenarios, approaches, challenges and perspectives. *arXiv preprint arXiv:2503.13415* (2025).
- [40] Rafflesia Khan and Nafiul Islam Khan. 2026. The cognitive companion: a lightweight parallel monitoring architecture for detecting and recovering from reasoning degradation in LLM agents. *arXiv preprint arXiv:2604.13759* (2026).
- [41] Rudolf Laine, Bilal Chughtai, Jan Betley, Kaivalya Hariharan, Jeremy Scheurer, Mikita Balesni, Marius Hobbhahn, Alexander Meinke, and Owain Evans. 2024. Me, myself, and ai: The situational awareness dataset (sad) for llms. *Advances in Neural Information Processing Systems* 37 (2024), 64010–64118.
- [42] Chingkwun Lam, Jiaxin Li, Lingfei Zhang, and Kuo Zhao. 2026. Governing evolving memory in LLM agents: Risks, mechanisms, and the stability and safety governed memory (SSGM) framework. *arXiv preprint arXiv:2603.11768* (2026).
- [43] Seojin Lee, ByeongJeong Kim, and Hwanhee Lee. 2026. Mitigating Cognitive Inertia in Large Reasoning Models via Latent Spike Steering. *arXiv preprint arXiv:2601.22484* (2026).
- [44] Ao Li, Yuexiang Xie, Songze Li, Fugee Tsung, Bolin Ding, and Yaliang Li. 2025. Agent-oriented planning in multi-agent systems. In *International Conference on Learning Representations*, Vol. 2025. 19495–19517.
- [45] Changmao Li and Jeffrey Flanigan. 2024. Rac: Efficient llm factuality correction with retrieval augmentation. *arXiv preprint arXiv:2410.15667* (2024).
- [46] Jing-Jing Li, Jianfeng He, Chao Shang, Devang Kulshreshtha, Xun Xian, Yi Zhang, Hang Su, Sandesh Swamy, and Yanjun Qi. 2025. STAC: When Innocent Tools Form Dangerous Chains to Jailbreak LLM Agents. *arXiv preprint arXiv:2509.25624* (2025).
- [47] Mengfei Liang, Archish Arun, Zekun Wu, Cristian Munoz, Jonathan Lutch, Emre Kazim, Adriano Koshiyama, and Philip Treleaven. 2024. Thames: An end-to-end tool for hallucination mitigation and evaluation in large language models. *arXiv preprint arXiv:2409.11353* (2024).
- [48] Yuxin Liang, Zhuoyang Song, Hao Wang, and Jiaying Zhang. 2024. Learning to trust your feelings: Leveraging self-awareness in llms for hallucination mitigation. In *Proceedings of the 3rd Workshop on Knowledge Augmented Methods for NLP*. 44–58.
- [49] Junda Lin, Zhaomeng Zhou, Zhi Zheng, Shuo Chen, Tong Xu, Yong Chen, and Enhong Chen. 2026. VIGIL: Defending LLM Agents Against Tool Stream Injection via Verify-Before-Commit. *arXiv preprint arXiv:2601.05755* (2026).
- [50] Xixun Lin, Yang Liu, Yancheng Chen, Yongxuan Wu, Yucheng Ning, Yilong Liu, Nan Sun, Shun Zhang, Bin Chong, Chuan Zhou, et al. 2026. SafeHarness: Lifecycle-Integrated Security Architecture for LLM-based Agent Deployment. *arXiv preprint arXiv:2604.13630* (2026).
- [51] Xixun Lin, Yucheng Ning, Jingwen Zhang, Yan Dong, Yilong Liu, Yongxuan Wu, Xiaohua Qi, Nan Sun, Yanmin Shang, Kun Wang, et al. 2025. Llm-based agents suffer from hallucinations: A survey of taxonomy, methods, and directions. *arXiv preprint arXiv:2509.18970* (2025).
- [52] Hongrui Liu, Duo Xu, Qianli Ma, Shuyang Xu, and Dong Qiu. 2026. Memory Poisoning Propagation and Repair Mechanism in Multi-Agent Collaborative Environments. In *Proceedings of the 2nd International Conference on Artificial Intelligence, Digital Media Technology and Social Computing*. 225–231.
- [53] Jiadi Liu, Ting Li, Qu Yuan Wang, Ying Wang, Zhiwei Guo, and Keping Yu. 2025. Unleashing collaborative potentials: Multifaceted collaboration among agents in multitask Internet of Things networks. *IEEE Internet of Things Journal* (2025).
- [54] Xinyun Liu, Zelei Cheng, Haodong Zhao, and Ronghua Xu. 2026. Security of Large Model-based Agents: A Survey on Adversarial, Poisoning, and Backdoor Attacks. (2026).
- [55] Xuannan Liu, Xiao Yang, Zekun Li, Peipei Li, and Ran He. 2026. AgentHallu: Benchmarking Automated Hallucination Attribution of LLM-based Agents. *arXiv preprint arXiv:2601.06818* (2026).
- [56] Jinghao Luo, Yuchen Tian, Chuxue Cao, Ziyang Luo, Hongzhan Lin, Kaixin Li, Chuyi Kong, Ruichao Yang, and Jing Ma. 2026. From Storage to Experience: A Survey on the Evolution of LLM Agent Memory Mechanisms. (2026).
- [57] Yibo Lyu, Gongwei Chen, Rui Shao, Weili Guan, and Liqiang Nie. 2026. PersonalAlign: Hierarchical Implicit Intent Alignment for Personalized GUI Agent with Long-Term User-Centric Records. *arXiv preprint arXiv:2601.09636* (2026).
- [58] Xinbei Ma, Ruotian Ma, Xingyu Chen, Zhengliang Shi, Mengru Wang, Jen-tse Huang, Qu Yang, Wenxuan Wang, Fanghua Ye, Qingxuan Jiang, et al. 2025. The Hunger Game Debate: On the Emergence of Over-Competition in Multi-Agent Systems. *arXiv preprint arXiv:2509.26126* (2025).
- [59] Utsav Maskey, Sumit Yadav, Mark Dras, and Usman Naseem. 2025. Safeconstellations: Steering llm safety to reduce over-refusals through task-specific trajectory. *arXiv preprint arXiv:2508.11290* (2025).
- [60] Ivan Milev, Mislav Balunović, Maximilian Baader, and Martin Vechev. 2025. ToolFuzz—Automated Agent Tool Testing. *arXiv preprint arXiv:2503.04479* (2025).
- [61] Scott Monteith, Tasha Glenn, John R Geddes, Peter C Whybrow, Eric Achtyes, and Michael Bauer. 2024. Artificial intelligence and increasing misinformation. *The British Journal of Psychiatry* 224, 2 (2024), 33–35.

- [62] Yutao Mou, Zhangchi Xue, Lijun Li, Peiyang Liu, Shikun Zhang, Wei Ye, and Jing Shao. 2026. ToolSafe: Enhancing Tool Invocation Safety of LLM-based agents via Proactive Step-level Guardrail and Feedback. *arXiv preprint arXiv:2601.10156* (2026).
- [63] Sadeeq Muhammad, Hussain Maharaz Yusuf, and Sandeep Kumar. 2026. Threats and Defenses in Large Language Models: A Review of Adversarial and Model Poisoning Techniques. (2026).
- [64] Akshat Naik, Patrick Quinn, Guillermo Bosch, Emma Gouné, Francisco Javier Campos Zabala, Jason Ross Brown, and Edward James Young. 2025. Agentmisalignment: Measuring the propensity for misaligned behaviour in llm-based agents. *arXiv preprint arXiv:2506.04018* (2025).
- [65] Vineeth Sai Narajala, Ken Huang, and Idan Habler. 2026. Securing genai multi-agent systems against tool squatting: A zero trust registry-based approach. In *2026 International Conference on AI x Data and Knowledge Engineering (AIDKE)*. IEEE, 91–93.
- [66] Elliot Nelson, Georgios Kollias, Payel Das, Subhjit Chaudhury, and Soham Dan. 2024. Needle in the haystack for memory based large language models. *arXiv preprint arXiv:2407.01437* (2024).
- [67] Zeinab Nezami, Shehr Bano, Abdelaziz Salama, Maryam Hafeez, and Syed Ali Raza Zaidi. 2025. Safety and Risk Pathways in Cooperative Generative Multi-Agent Systems: A Telecom Perspective. In *Proceedings of the 2nd ACM Workshop on Open and AI RAN*. 50–55.
- [68] Tam Nguyen, Moses Ndebugre, and Dheeraj Arremsetty. 2026. Security Considerations for Multi-agent Systems. *arXiv preprint arXiv:2603.09002* (2026).
- [69] Kai Tzu-iunn Ong, Namyoun Kim, Minju Gwak, Hyungjoo Chae, Taeyoon Kwon, Yohan Jo, Seung-won Hwang, Dongha Lee, and Jinyoung Yeo. 2025. Towards lifelong dialogue agents via timeline-based memory management. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*. 8631–8661.
- [70] Deonna M Owens, Ryan Rossi, Sungchul Kim, Tong Yu, Franck Dernoncourt, Xiang Chen, Ruiyi Zhang, Jiuxiang Gu, Hanieh Deilamsalehy, and Nedim Lipka. 2025. Multi-LLM Debiasing Framework. In *Proceedings of the 15th International Conference on Recent Advances in Natural Language Processing-Natural Language Processing in the Generative AI Era*. 843–853.
- [71] Wenbo Pan, Shujie Liu, Xiangyang Zhou, Shiwei Zhang, Wanlu Shi, Mirror Xu, and Xiaohua Jia. 2026. M★: Every Task Deserves Its Own Memory Harness. *arXiv preprint arXiv:2604.11811* (2026).
- [72] Xiaoyu Pan, Yang Bai, Ke Zou, Yang Zhou, Jun Zhou, Huazhu Fu, Yih-Chung Tham, and Yong Liu. 2025. EH-Benchmark: Ophthalmic hallucination benchmark and agent-driven top-down traceable reasoning workflow. *Information Fusion* (2025), 103631.
- [73] Jinhu Qi, Muzhi Li, Jiahong Liu, Yuqin Shu, Dianzhi Yu, Shicheng Ma, Wenqian Cui, Yiyang Zhao, Yiyi Chen, Ruoxi Jiang, et al. 2026. Towards trustworthy agentic AI: a comprehensive survey of safety, robustness, privacy, and system security. *Academia AI and Applications* 2, 2 (2026).
- [74] Cheng Qian, Emre Can Acikgoz, Hongru Wang, Xiusi Chen, Avirup Sil, Dilek Hakkani-Tur, Gokhan Tur, and Heng Ji. 2025. SMART: Self-aware agent for tool overuse mitigation. In *Findings of the Association for Computational Linguistics: ACL 2025*. 4604–4621.
- [75] Preston Rasmussen, Pavlo Paliychuk, Travis Beauvais, Jack Ryan, and Daniel Chalef. 2025. Zep: a temporal knowledge graph architecture for agent memory. *arXiv preprint arXiv:2501.13956* (2025).
- [76] Abhishek Rath. 2026. Agent Drift: Quantifying Behavioral Degradation in Multi-Agent LLM Systems Over Extended Interactions. *arXiv preprint arXiv:2601.04170* (2026).
- [77] Adam Roe, Samuel Richardson, Joseph Schneider, Anthony Cummings, Nicholas Forsberg, and Jonathan Klein. 2024. Semantic drift mitigation in large language model knowledge retention using the residual knowledge stability concept. *Authorea Preprints* (2024).
- [78] Charles Schepanowski and Charles Ling. 2025. On the limits of innate planning in large language models. *arXiv preprint arXiv:2511.21591* (2025).
- [79] Standard Self-Reflection. 2024. Self-contrast: Better reflection through inconsistent solving perspectives. (2024).
- [80] Shuai Shao, Qihan Ren, Chen Qian, Boyi Wei, Dadi Guo, Jingyi Yang, Xinhao Song, Linfeng Zhang, Weinan Zhang, Dongrui Liu, et al. 2025. Your agent may misevolve: Emergent risks in self-evolving llm agents. *arXiv preprint arXiv:2509.26354* (2025).
- [81] Saksham Sahai Srivastava and Haoyu He. 2025. MemoryGraft: Persistent compromise of LLM agents via poisoned experience retrieval. *arXiv preprint arXiv:2512.16962* (2025).
- [82] Hang Su, Jun Luo, Chang Liu, Xiao Yang, Yichi Zhang, Yinpeng Dong, and Jun Zhu. 2026. A survey on autonomy-induced security risks in large model-based agents. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2026).
- [83] Chuxiong Sun, Peng He, Qirui Ji, Zehua Zang, Jiangmeng Li, Rui Wang, and Wei Wang. 2026. M2i2: Learning efficient multi-agent communication via masked state modeling and intention inference. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 40. 25672–25681.
- [84] Luoxi Tang, Rupali Rajendra Vaje, Yuqiao Meng, Sakshi Sunil Narkar, Weicheng Ma, Zeyu Ding, Dazheng Zhang, and Zhaohan Xi. 2026. The Trap of Trajectory: Towards Understanding and Mitigating Spurious Correlations in Agentic Memory. *arXiv preprint arXiv:2605.09330* (2026).
- [85] Ruixiang Tang, Dehan Kong, Longtao Huang, et al. 2023. Large language models can be lazy learners: Analyze shortcuts in in-context learning. In *Findings of the association for computational linguistics: ACL 2023*. 4645–4657.

- [86] Yaxin Tang, Yijia Liu, Jiahe Lan, Zheng Yan, and Erol Gelenbe. 2025. Security of LLM-based Agents Regarding Attacks, Defenses, and Applications: A Comprehensive Survey. *Information Fusion* (2025), 103941.
- [87] Rizwan Tanveer. 2026. Multi-Agent Orchestration: Coordination, Trust, and Cascading Failures. *Trust, and Cascading Failures (May 08, 2026)* (2026).
- [88] Zhen Tao, Jinxiang Zhao, Peng Liu, Dinghao Xi, Yanfang Chen, Wei Xu, and Zhiyu Li. 2026. MemConflict: Evaluating Long-Term Memory Systems Under Memory Conflicts. *arXiv preprint arXiv:2605.20926* (2026).
- [89] Deemah H Tashman and Soumaya Cherkaoui. 2026. Trustworthy AI-Driven Dynamic Hybrid RIS: Joint Optimization and Reward Poisoning-Resilient Control in Cognitive MISO Networks. *IEEE Transactions on Network and Service Management* (2026).
- [90] Khanh-Tung Tran, Dung Dao, Minh-Duong Nguyen, Quoc-Viet Pham, Barry O'Sullivan, and Hoang D Nguyen. 2025. Multi-agent collaboration mechanisms: A survey of llms. *arXiv preprint arXiv:2501.06322* (2025).
- [91] Ken Tsui. 2025. Self-correction bench: Uncovering and addressing the self-correction blind spot in large language models. *arXiv preprint arXiv:2507.02778* (2025).
- [92] Tu Vu, Mohit Iyyer, Xuezhi Wang, Noah Constant, Jerry Wei, Jason Wei, Chris Tar, Yun-Hsuan Sung, Denny Zhou, Quoc Le, et al. 2024. Freshllms: Refreshing large language models with search engine augmentation. In *Findings of the Association for Computational Linguistics: ACL 2024*. 13697–13720.
- [93] Feiyang Wang, Hengguang Cui, Linghao Yang, Chi Shing Lee, Zhongkang Li, and Chenfeiyu Wen. 2026. Detecting and repairing role drift in multi-agent collaboration with lightweight protocols. In *2026 International Conference on Communication Networks and Machine Learning (CNML)*. IEEE, 1420–1424.
- [94] Hongru Wang, Cheng Qian, Manling Li, Jiahao Qiu, Boyang Xue, Mengdi Wang, Heng Ji, Amos Storkey, and Kam-Fai Wong. 2025. Position: Agent Should Invoke External Tools ONLY When Epistemically Necessary. *arXiv preprint arXiv:2506.00886* (2025).
- [95] Jiacheng Wang, Jinchang Hou, Fabian Wang, Ping Jian, Chenfu Bao, and Zhonghou Lv. 2026. HINTBench: Horizon-agent Intrinsic Non-attack Trajectory Benchmark. *arXiv preprint arXiv:2604.13954* (2026).
- [96] Kun Wang, Guibin Zhang, Zhenhong Zhou, Jiahao Wu, Miao Yu, Shiqian Zhao, Chenlong Yin, Jinhu Fu, Yibo Yan, Hanjun Luo, et al. 2025. A comprehensive survey in llm (-agent) full stack safety: Data, training and deployment. *arXiv preprint arXiv:2504.15585* (2025).
- [97] Sion Weatherhead, Flora Salim, and Aaron Belbasis. 2025. Illusions of reflection: open-ended task reveals systematic failures in Large Language Models' reflective reasoning. *arXiv preprint arXiv:2510.18254* (2025).
- [98] Qianshan Wei, Tengchao Yang, Yaochen Wang, Xinfeng Li, Lijun Li, Zhenfei Yin, Yi Zhan, Thorsten Holz, Zhiqiang Lin, and XiaoFeng Wang. 2025. A-memguard: A proactive defense framework for llm-based agent memory. *arXiv preprint arXiv:2510.02373* (2025).
- [99] Junlin Wu, Jiong Xiao Wang, Chaowei Xiao, Chenguang Wang, Ning Zhang, and Yevgeniy Vorobeychik. 2025. Preference poisoning attacks on reward model learning. In *2025 IEEE Symposium on Security and Privacy (SP)*. IEEE, 1622–1640.
- [100] Weimin Xiong, Ke Wang, Yifan Song, Hanchao Liu, Sai Zhou, Wei Peng, and Sujian Li. 2025. More Vulnerable than You Think: On the Stability of Tool-Integrated LLM Agents. *arXiv preprint arXiv:2506.21967* (2025).
- [101] Binyan Xu, Xilin Dai, and Kehuan Zhang. 2026. Contextual Agentic Memory is a Memo, Not True Memory. *arXiv preprint arXiv:2604.27707* (2026).
- [102] Tianyang Xu, Shujin Wu, Shizhe Diao, Xiaozhe Liu, Xingyao Wang, Yangyi Chen, and Jing Gao. 2024. Sayself: Teaching llms to express confidence with self-reflective rationales. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. 5985–5998.
- [103] Xiaoyu Xu, Minxin Du, Qipeng Xie, Haobin Ke, Qingqing Ye, and Haibo Hu. 2026. When Routine Chats Turn Toxic: Unintended Long-Term State Poisoning in Personalized Agents. *arXiv preprint arXiv:2605.06731* (2026).
- [104] Zhenjie Xu, Wenqing Chen, Yi Tang, Xuanying Li, Cheng Hu, Zhixuan Chu, Kui Ren, Zibin Zheng, and Zhichao Lu. 2025. Mitigating social bias in large language models: A multi-objective approach within a multi-agent framework. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39. 25579–25587.
- [105] Zijie Xu, Minfeng Qi, Shiqing Wu, Lefeng Zhang, Qiwen Wei, Han He, and Ningran Li. 2026. The trust paradox in llm-based multi-agent systems: When collaboration becomes a security vulnerability. *IEEE Transactions on Computational Social Systems* (2026).
- [106] Zhenlin Xu, Xiaogang Zhu, Yu Yao, Minhui Xue, and Yiliao Song. 2026. From storage to steering: Memory control flow attacks on LLM agents. *arXiv preprint arXiv:2603.15125* (2026).
- [107] Shilin Yan, Jintao Tong, Hongwei Xue, Xiaojun Tang, Yangyang Wang, Kunyu Shi, Guannan Zhang, Ruixuan Li, and Yixiong Zou. 2026. Act wisely: Cultivating meta-cognitive tool use in agentic multimodal models. *arXiv preprint arXiv:2604.08545* (2026).
- [108] Sikuan Yan, Xiufeng Yang, Zuchao Huang, Ercong Nie, Zifeng Ding, Zonggen Li, Xiaowen Ma, Jinhe Bi, Kristian Kersting, Jeff Z Pan, et al. 2025. Memory-r1: Enhancing large language model agents to manage and utilize memories via reinforcement learning. *arXiv preprint arXiv:2508.19828* (2025).
- [109] Wenhao Yan, Ning An, Linxu Li, Bingsheng Bi, Bo Jiang, Zhigang Lu, Baoxu Liu, Junrong Liu, and Cong Dong. 2026. ProvAgent: Threat Detection Based on Identity-Behavior Binding and Multi-Agent Collaborative Attack Investigation. *arXiv preprint arXiv:2603.09358* (2026).

- [110] Ruihan Yang, Fanghua Ye, Xiang We, Ruoqing Zhao, Kang Luo, Xinbo Xu, Bo Zhao, Ruotian Ma, Shanyi Wang, Zhaopeng Tu, et al. 2026. Think Fast and Slow: Step-Level Cognitive Depth Adaptation for LLM Agents. *arXiv preprint arXiv:2602.12662* (2026).
- [111] Tiankai Yang, Jiatae Li, Yi Nian, Shen Dong, Ruiyao Xu, Ryan Rossi, Kaize Ding, and Yue Zhao. 2026. No Attacker Needed: Unintentional Cross-User Contamination in Shared-State LLM Agents. *arXiv preprint arXiv:2604.01350* (2026).
- [112] Yingxuan Yang, Huacan Chai, Shuai Shao, Yuanyi Song, Siyuan Qi, Renting Rui, and Weinan Zhang. 2026. Agentnet: Decentralized evolutionary coordination for llm-based multi-agent systems. *Advances in Neural Information Processing Systems* 38 (2026), 107309–107336.
- [113] Yingxuan Yang, Mulei Ma, Yuxuan Huang, Huacan Chai, Chenyu Gong, Haoran Geng, Yuanjian Zhou, Ying Wen, Meng Fang, Muhao Chen, et al. 2025. Agentic web: Weaving the next web with ai agents (2025). URL <https://arxiv.org/abs/2507.21206> (2025).
- [114] Junjie Ye, Sixian Li, Guanyu Li, Caishuang Huang, Songyang Gao, Yilong Wu, Qi Zhang, Tao Gui, and Xuan-Jing Huang. 2024. Toolsword: Unveiling safety issues of large language models in tool learning across three stages. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 2181–2211.
- [115] Weibin Ye, Hanfu Wang, and Weidong Chen. 2025. Path Coordination for Robust, Fast and Scalable Multi-Agent Path Finding under Unforeseen Delays. *IEEE Transactions on Automation Science and Engineering* (2025).
- [116] Peng Yifeng, Zhizheng Wu, and Chen Chen. 2025. Swallowing the Poison Pills: Insights from Vulnerability Disparity Among LLMs. In *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*. 38–55.
- [117] Bo Yin, Qi Li, and Xinchao Wang. 2026. On-Policy Self-Evolution via Failure Trajectories for Agentic Safety Alignment. *arXiv preprint arXiv:2605.11882* (2026).
- [118] Shasha Yu, Fiona Carroll, and Barry L Bentley. 2026. The Causal Impact of Tool Affordance on Safety Alignment in LLM Agents. *arXiv preprint arXiv:2603.20320* (2026).
- [119] Chaoyun Zhang, Shilin He, Jiaxu Qian, Bowen Li, Liquan Li, Si Qin, Yu Kang, Minghua Ma, Guyue Liu, Qingwei Lin, et al. 2024. Large language model-brained gui agents: A survey. *arXiv preprint arXiv:2411.18279* (2024).
- [120] Dylan Zhang, Yanshan Lin, Zhengkun Wu, Yihang Sun, Bingxuan Li, Dianqi Li, and Hao Peng. 2026. Useful Memories Become Faulty When Continuously Updated by LLMs. *arXiv preprint arXiv:2605.12978* (2026).
- [121] Jinchuan Zhang, Lu Yin, Yan Zhou, and Songlin Hu. 2025. Agentalign: Navigating safety alignment in the shift from informative to agentic large language models. *arXiv preprint arXiv:2505.23020* (2025).
- [122] Yixiang Zhang, Xinhao Deng, Jiaqing Wu, Yue Xiao, Ke Xu, and Qi Li. 2026. AgentWard: A Lifecycle Security Architecture for Autonomous AI Agents. *arXiv preprint arXiv:2604.24657* (2026).
- [123] Yuxuan Zhang, Yangyang Feng, Daifeng Li, Kexin Zhang, Junlan Chen, and Bowen Deng. 2025. Can Competition Enhance the Proficiency of Agents Powered by Large Language Models in the Realm of News-driven Time Series Forecasting? *arXiv preprint arXiv:2504.10210* (2025).
- [124] Yechao Zhang, Shiqian Zhao, Jie Zhang, Gelei Deng, Jiawen Zhang, Xiaogeng Liu, Chaowei Xiao, and Tianwei Zhang. 2026. Mind your heartbeat! claw background execution inherently enables silent memory pollution. *arXiv preprint arXiv:2603.23064* (2026).
- [125] Zeyu Zhang, Quanyu Dai, Xiaohe Bo, Chen Ma, Rui Li, Xu Chen, Jieming Zhu, Zhenhua Dong, and Ji-Rong Wen. 2025. A survey on the memory mechanism of large language model-based agents. *ACM Transactions on Information Systems* 43, 6 (2025), 1–47.
- [126] Xufeng Zhao, Mengdi Li, Wenhao Lu, Cornelius Weber, Jae-Hee Lee, Kun Chu, and Stefan Wermter. 2024. Enhancing zero-shot chain-of-thought reasoning in large language models through logic. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*. 6144–6166.
- [127] Zhi Zheng, Qian Feng, Hang Li, Alois Knoll, and Jianxiang Feng. 2024. Evaluating uncertainty-based failure detection for closed-loop LLM planners. *arXiv preprint arXiv:2406.00430* (2024).
- [128] Chenyu Zhou, Huacan Chai, Wenteng Chen, Zihan Guo, Rong Shan, Yuanyi Song, Tianyi Xu, Yingxuan Yang, Aofan Yu, Weiming Zhang, et al. 2026. Externalization in LLM Agents: A Unified Review of Memory, Skills, Protocols and Harness Engineering. *arXiv preprint arXiv:2604.08224* (2026).
- [129] Yifan Zhu, Lijia Yu, and Xiao-Shan Gao. 2026. Provable watermarking for data poisoning attacks. *Advances in Neural Information Processing Systems* 38 (2026), 170840–170881.

Received 20 February 2026; revised 12 March 2026; accepted 5 June 2026